

Name _____ Date _____ Period _____

Graphing $f(x) = |x|$ and Key Attributes

Algebra II Honors | Unit 2 | Day 1 | TEK A2.2.A

Objective: I can graph $f(x) = |x|$ from its piecewise definition, derive its key attributes algebraically, and compare its behavior to a linear function.

Vocabulary

Piecewise function — a function defined by _____ on different parts of the domain.

Absolute value, formally — $|x| = x$ if $x \geq 0$, and $|x| = \underline{\hspace{1cm}}$ if $x < 0$.

Even function — $f(-x) = f(x)$ for every x in the domain; the graph is symmetric about the _____.

Vertex — where the two pieces meet: _____ for $f(x) = |x|$.

Part 1 — The Piecewise Definition

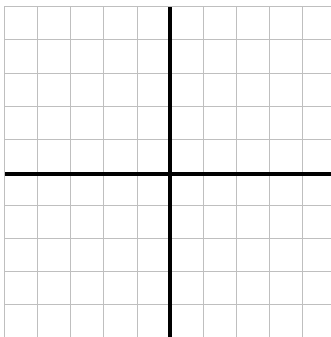
$$f(x) = |x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0 \end{cases}$$

Evaluate using the correct piece.

$$f(-4) = \underline{\hspace{1cm}} \text{ (uses } -x) \quad f(6) = \underline{\hspace{1cm}} \text{ (uses } x) \quad f(0) = \underline{\hspace{1cm}} \quad f(-0.5) = \underline{\hspace{1cm}} \text{ (uses } -x)$$

Part 2 — Graph Each Piece

Graph $y = x$ for $x \geq 0$ and $y = -x$ for $x < 0$ on the SAME grid - two rays meeting at the vertex. Each square is 1 unit, both axes run -5 to 5.



Part 3 — Key Attributes, Derived From the Definition

Domain: _____ Vertex: _____

Derive the range from the two pieces, instead of just reading it off the graph:

For $x \geq 0$, the rule is $y = x$, so y takes every value _____

For $x < 0$, the rule is $y = -x$, so y takes every value _____ (since x is negative, $-x$ is positive)

Combining both pieces, the range is _____

Now show algebraically that $f(x) = |x|$ is an even function:

$f(-x) = |-x| = \underline{\hspace{1cm}} \rightarrow$ confirmed even, so the graph is symmetric about the _____

Now derive the x - and y -intercepts from the definition instead of reading them off the graph:

x -piece: $x = 0 \rightarrow x = \underline{\hspace{1cm}}$ $-x$ -piece: $-x = 0 \rightarrow x = \underline{\hspace{1cm}}$ (both agree)

So the only x -intercept is _____, which is also the y -intercept, since $f(0) = \underline{\hspace{1cm}}$

Part 4 — Compare and Explain

1. Explain why $f(x) = |x|$ has a minimum value but a line $y = mx + b$ ($m \neq 0$) does not.

Sample: _____

2. For which values of x does $|x| = x$ (the two pieces agree)? _____
3. For which values of x does $|x| = -x$? _____ (at $x = 0$ both pieces give the same value, 0)
4. Is $f(x) = |x|$ a function? Justify using the vertical line test.

Yes _____

Part 5 — Real-World: Distance, Justified

A hiker's position is x miles from a trailhead at 0 (negative = west, positive = east). Distance from the trailhead is $d(x) = |x|$.

Using the piecewise definition, explain why $d(x)$ is never negative, no matter which direction the hiker walks.

Sample: _____

Part 6 — Going Further

1. For which value(s) of x does $|x| = 5$? _____

Explain in one sentence why there are two answers.

Sample: _____

2. For which value(s) of x does $|x| = -3$? _____

Explain why, using the range you derived in Part 3.

Sample: _____

3. Could $|x|$ ever equal a number strictly between -1 and 0, like -0.5? Explain.

No _____

Exit Ticket

Justify algebraically that $f(x) = |x|$ is symmetric about the y -axis, then give its domain, range, and vertex.

Domain _____ Range _____ Vertex _____